Why do we need GIS?

GIS (Geographic Information System) is a technology used to visualize, analyze, and manage spatial or geographic data.

QGIS is a free and open-source GIS software that allows us to create maps, analyze spatial data, and represent georeferenced information.

We are going to use QGIS to:

- Load GPS points
- Represent measurement profiles (e.g., from fieldwork or lab)
- Visualize and export maps with scale bars, north arrows, and labels
- Combine spatial data with analysis tools

Field Data

GPS points, survey transects, and measurement locations need to be placed in geographic context.

Spatial Distribution

Visualize how measured variables (e.g. resistivity, GPR depth) vary across the study area.

Multiple Layers

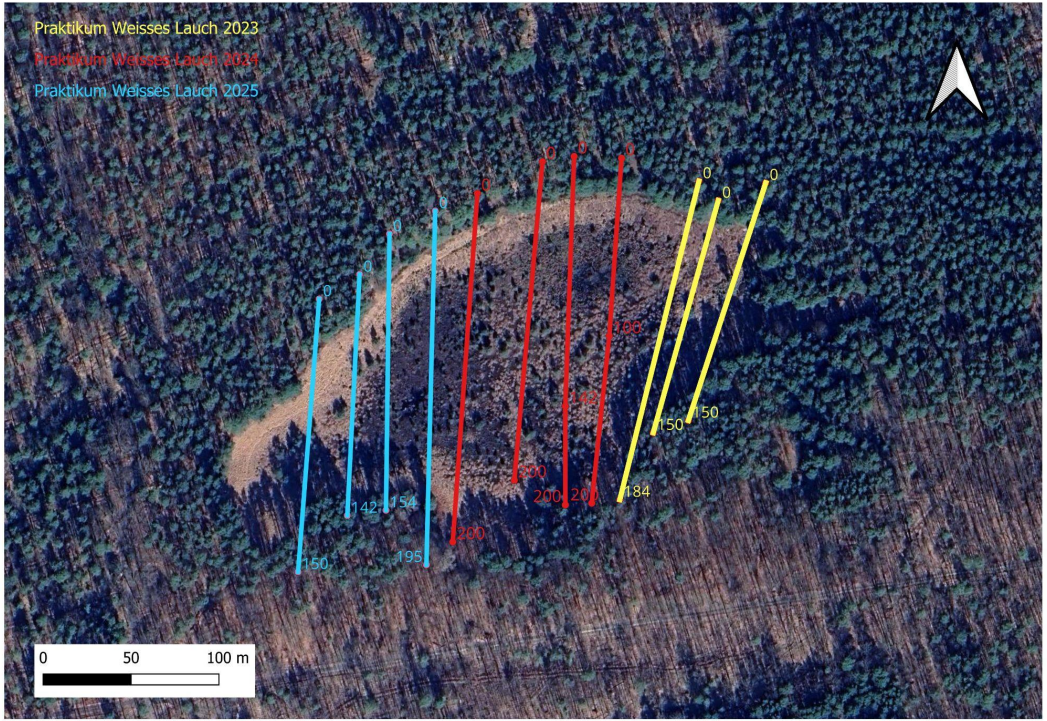
Overlay geology, land use, topography, or satellite imagery with your geophysical datasets.

Map Production

Export publication-ready maps with scale bars, north arrows, legends, and proper projections.

Why map measurement values over a basemap?

Example 1 — EM/ERT survey, Weisses Lauch



Each field measurement (resistivity, conductivity, TMI…) has a geographic location.
Loading those values into QGIS and displaying them over a basemap reveals spatial patterns: which zones are anomalous, where to focus next.

EM → conductivity (x)

Why map measurement values over a basemap?

Example 2 — Magnetometry, BTU Campus Nord

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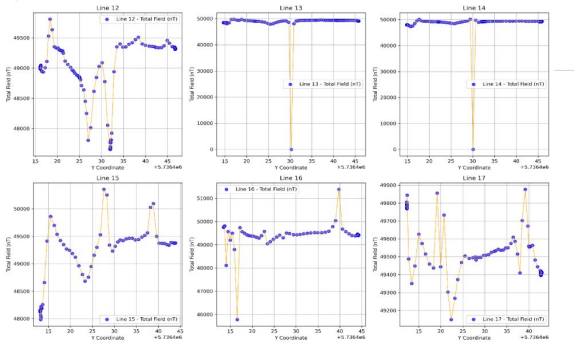
Magnetometer ⟶ Total Magnetic Intensity (x)



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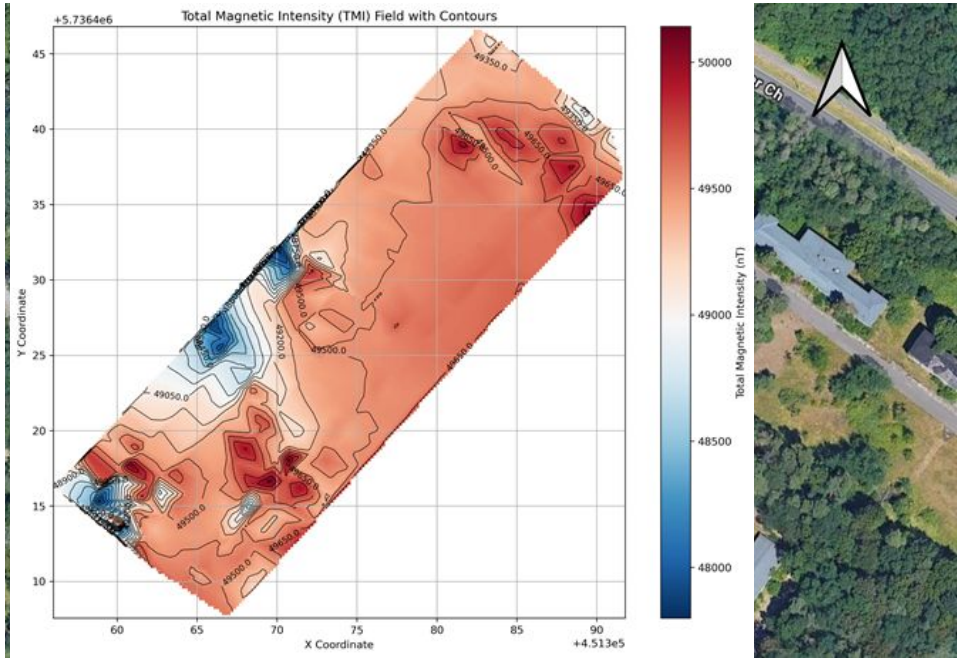

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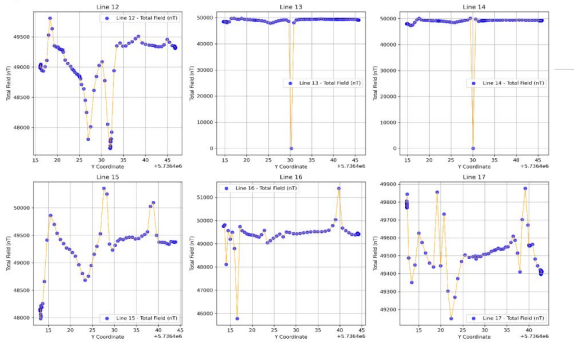

Types of GIS data

VECTOR

- Points — GPS measurements, sampling sites
- Lines — Survey transects, roads, rivers
- Polygons — Study area, geological units

You will mainly work with POINTS and LINES in this course.

RASTER

- Grid of cells (pixels), each with a value
- Satellite imagery, DEMs, interpolated maps
- Continuous data: elevation, temperature...

Useful for background maps and interpolated geophysical results.